

DentalCare Pro Enterprise — Release Notes

Version 1.0.0 Production Release | Enterprise Dental Practice Management System

DentalCare Pro Enterprise v1.0.0 is the official commercial production release designed for solo, group, and multi-operatory dental practices. This release delivers a fully integrated, offline-capable desktop environment with zero cloud dependency requirements, combining high-performance FastAPI backends, modern interactive odontograms, and HIPAA-certified security.

Major Highlights & Clinical Capabilities

- **Interactive Dental Odontogram:** Full 2D/3D dental chart supporting both FDI Two-Digit and Universal Numbering systems. Features anatomical surface-level charting (mesial, occlusal, distal, buccal, lingual), visual color-coded restorations, crowns, implants, endodontic canals, and extractions.
- **Electronic Health Records (EHR):** Comprehensive medical histories, periodontal pocket depth probing (6-point charting with bleeding on probing indicators), clinical progress notes, and allergies tracking with automated drug interaction alerts.
- **Multi-Operatory Scheduling:** Drag-and-drop chairside appointment calendar with operatory conflict detection, dentist availability rules, automated appointment reminders, and recall intervals.
- **Digital Radiography & DICOM Viewer:** Native DICOM 3.0 medical imaging engine supporting panoramic, periapical, bitewing, and CBCT scans with window/level contrast adjustment, measurement calibrations, and annotations.
- **Enterprise Billing & Insurance:** Automated treatment fee estimation, standard dental procedure codes (CDT / ADA), electronic insurance claim preparation (EDI 837D format), patient ledger, and split payment allocation.
- **Desktop Application Shell:** Packaged with Microsoft Edge WebView2 engine for zero-browser-tab clutter, automatic backend process supervision, crash recovery, and offline data caching.

System Requirements & Compatibility Matrix

Component	Minimum Specification	Recommended Specification
Operating System	Windows 10 64-bit (Build 19041+)	Windows 11 64-bit Pro / Enterprise
Processor	Intel Core i3 / AMD Ryzen 3 (2.0 GHz)	Intel Core i5 / AMD Ryzen 5 (3.0 GHz+)
Memory (RAM)	4 GB DDR4	8 GB - 16 GB DDR4/DDR5
Storage	500 MB free disk space	2 GB+ SSD for imaging & local backups
Database Engine	PostgreSQL 14.x / 15.x / 16.x / 17.x	PostgreSQL 16+ on LAN Server / Localhost
Display Resolution	1280 x 768 pixels	1920 x 1080 (Full HD) chairside display

Security & Verification Sign-Off

DentalCare Pro v1.0.0 has passed 249 unit, integration, and security test suites with 100% pass rate. SHA-256 cryptographic hashes for every binary file in this release are documented in **Checksums.txt**.